June 2026

1st Krones

Vision AI challenge

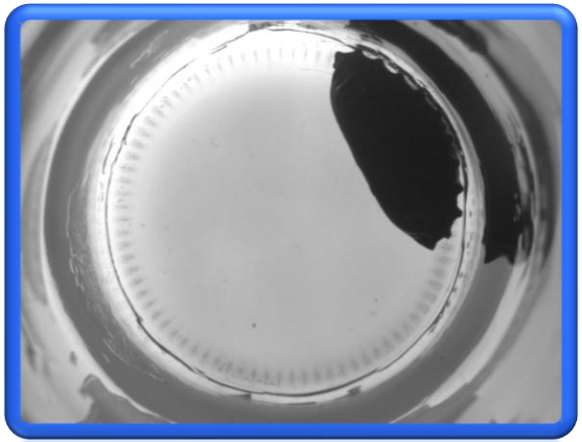


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Bottle reusability classification ·
EfficientNet-B3 ·
PyTorch ·

THE PROBLEM

Returnable bottles need inspection before refilling.



The challenge

High-speed production lines can't rely on manual checks — too slow, too inconsistent. Krones' Linatronic AI uses deep learning to automate this at scale.

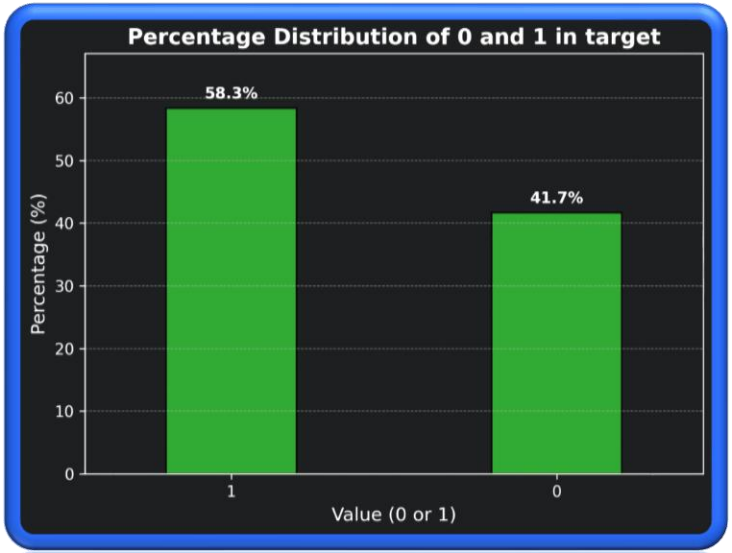
Our task

Binary classification of every bottle image:

- Reusable — safe to refill
- Not reusable — reject

DATASET

35,342 training images.
4,418 test images.



80 / 20
Train / validation split

384 × 384
Input resolution (px)

COCO JSON
Annotation format

Preprocessing: ROI crop from COCO bounding box → Resize → Flip + rotation augmentation → ImageNet normalisation

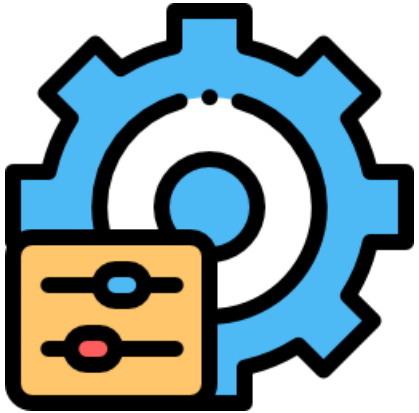
OUR APPROACH

Transfer learning on EfficientNet-B3

01	**ROI Crop —**	Parsing COCO JSON (*category_id = 22*) → cropping each image to the bottle bounding box. Removes all irrelevant background.
02	**Augment —**	*RandomAdjustSharpness*, *RandomAutocontrast*, *RandomHorizontalFlip (p=0.5), RandomRotation* ±180°. Bottles arrive at any orientation on real production lines.
03	**EfficientNet-B3 —**	ImageNet pre-trained weights. Final classifier head replaced: in_features → 2 classes (reusable / not reusable). EfficientNet-B3 balances accuracy & efficiency via compound scaling.
04	**Train & save —**	AdamW (lr 0.0005, weight_decay 0.01), CosineAnnealing scheduler, Cross Entropy Loss. After trying Fast-RCNN (object detection), ConvNeXt and Resnet18, EfficientNet-B3 delivered the best results with high performance and fewer computational resources.

TRAINING CONFIG

Simple setup
Consistent results



Batch size	**32**	Optimizer	**AdamW**
Epochs	**up to 15**	Scheduler	**CosineAnnealing**
Learning rate	**0.0005**	Loss	**Cross Entropy Loss**

RESULTS

Validation loss decreased throughout training.

Epochs	Train Loss	Valid Loss	LR	Accuracy	F1 (Macro)
1 / 15	0.2021	0.1586	0.000500	0.9311	0.9299
2 / 15	0.1478	0.1302	0.000491	0.9464	0.9446
3 / 15	0.1326	0.1129	0.000467	0.9542	0.9528
4 / 15	0.1215	0.1479	0.000427	0.9421	0.9410
5 / 15	0.1124	0.1190	0.000375	0.9523	0.9510
6 / 15	0.1031	0.1125	0.000315	0.9566	0.9550
7 / 15	0.0977	0.1097	0.000250	0.9581	0.9568
8 / 15	0.0880	0.1043	0.000185	0.9584	0.9571
9 / 15 ★	**0.0802**	**0.1013 ★**	**0.000125**	**0.9602**	**0.9589**
10 / 15	0.0743	0.1030	0.000073	0.9584	0.9571
11 / 15	0.0684	0.1079	0.000033	0.9604	0.9590
12 / 15	0.0650	0.1060	0.000009	0.9576	0.9562

Best model saved at epoch 9

val loss 0.1013

> 50% reduction from the initial training loss to the best validation loss

0.2021 → 0.1013

[As it can be expected, the other final epochs have worse results because of ***model overfitting*** on the training set]

WHY IT WORKS

Three design choices that matter.



01 **ROI Cropping removes noise.**

Without it, the model learns background patterns, not bottle defects. Cropping to the bounding box directs every gradient update to what counts.

02 **Rotation augmentation reflects reality.**

Bottles arrive in any orientation on a real production line. RandomRotation ±180° teaches the model to be rotation-invariant from day one.

03 **EfficientNet-B3 is efficient enough to matter.**

High performance with fewer resources — boosting the 30% efficiency score. Pretrained on 1.4M ImageNet images, it understands geometric shapes, edges, textures and curves for defect detection.

HOW IT EVOLVED

Efficient-Net (B3) really makes a difference!

ResNet-18 (early baseline)

At first – a solid, fast initial benchmark but lacked the parameter efficiency needed.

→

Fast-RCNN (Score: ~0.940)

Built for complex object detection. Since we already used COCO JSON to extract ROIs, the bounding-box overhead of RCNN was unnecessary and sub-optimal for binary classification.

→

ConvNeXt (Score: ~0.957)

Modern architecture, but extremely data-hungry. On our 35k image dataset, it converged slower and was more prone to overfitting

→

EfficientNet-B3 (final model score: 0.966*)

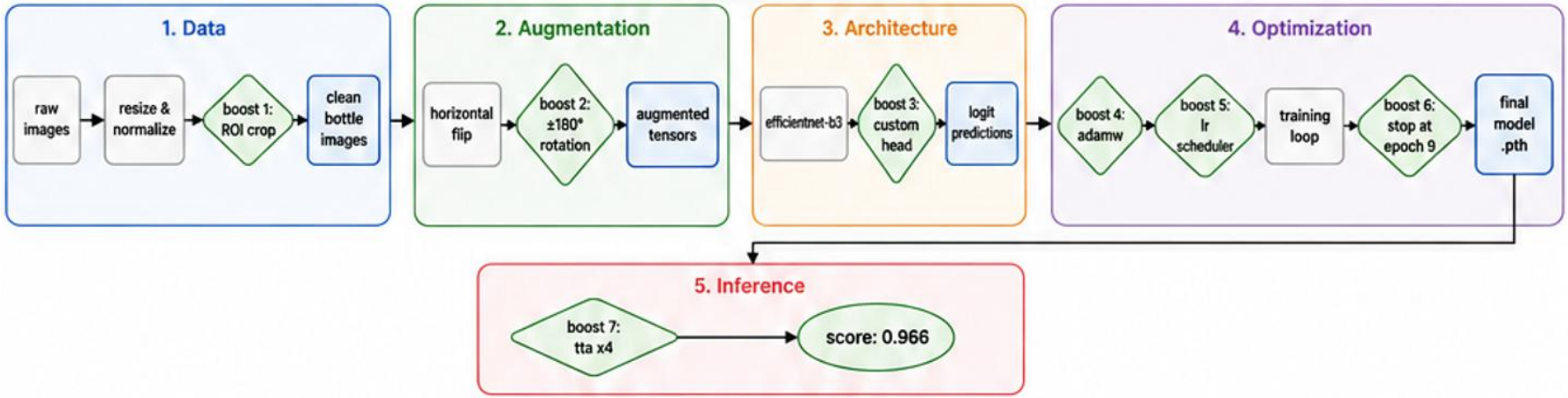
The optimal choice. Its "Compound scaling" seamlessly balances network depth, width, and resolution.

[Improvements: Adam to AdamW, 4 x rotational test time augmentation, more epochs included (7 → 10 → 15]

[*on the hidden test set]

CONCLUSION

A focused model built for a real problem.



- EfficientNet-B3 + transfer learning — from the strong baseline, fast inference
- ROI cropping focuses the model on what matters
- 9 epochs of consistent improvement — val loss **0.2021 → 0.1013**
- GPU-focused and optimized – cuda utilization, AMP, pin memory, parallelism
- Test time augmentation x4 — 0°/90°/180°/270° rotation ensemble at inference


Thank
you!